A

HSBTE LEET 2026 — Cheat Sheet

Basic Sciences (Q.1–25 | 25 Marks)

Sub-Sec	Subject	Qs	Marks
a-1	Mathematics	8	8
a-2	Physics	8	8
a-3	Chemistry	3	3
a-4	Communication Skills	3	3
a-5	General Awareness	3	3
TOTAL		**25**	**25**

a-1 MATHEMATICS (8 Questions | 8 Marks)

■ Sequences & Series

- Arithmetic Progressions (AP): nth term = $a+(n-1)d$, Sum = $n/2[2a+(n-1)d]$
- Binomial Expansion: $(a+b)^n$, general term $T(r+1) = nCr \cdot a^{(n-r)} \cdot b^r$

■ Algebra

- Complex Numbers: $z = a+ib$, $|z|$, conjugate, De Moivre's theorem
- Logarithm: log rules — $\log(mn)=\log m+\log n$, $\log(m/n)=\log m–\log n$, $\log m^n=n\cdot\log m$
- Permutation & Combination: $nPr = n!/(n–r)!$, $nCr = n!/r!(n–r)!$
- Matrices: Types, operations, determinant, inverse, rank

■ Probability

- $P(A)$ = favourable/total outcomes; $P(A\cup B) = P(A)+P(B)–P(A\cap B)$

■ Trigonometry

- $\sin^2\theta+\cos^2\theta=1$, $1+\tan^2\theta=\sec^2\theta$, $1+\cot^2\theta=\operatorname{cosec}^2\theta$
- Compound angles, multiple angles, sub-multiple angle formulas

■ Coordinate Geometry

- Straight Line: slope $m=(y_2–y_1)/(x_2–x_1)$, $y=mx+c$, distance formula
- Circle: $(x–h)^2+(y–k)^2=r^2$, general form $x^2+y^2+2gx+2fy+c=0$

■ Calculus

- Differentiation: standard derivatives, chain/product/quotient rules
- Successive Differentiation: Leibnitz theorem
- Maxima/Minima: $dy/dx=0$, check d^2y/dx^2 sign
- Integration: standard integrals, by-parts, substitution
- Definite Integral: limits a to b, properties
- Application: area under curve, differential equations (1st order 1st degree)

■ Statistics

- Mean, Mode, Median, Mean Deviation, Standard Deviation — all formulas

a-2 PHYSICS (8 Questions | 8 Marks)

- Units & Dimensions: SI units, dimensional analysis, limitations
- Motion (1D) & Newton's Laws: $v=u+at$, $s=ut+\frac{1}{2}at^2$, $v^2=u^2+2as$; $F=ma$; action-reaction
- Work, Energy & Conservation: $W=F\cdot d\cdot\cos\theta$; $KE=\frac{1}{2}mv^2$; $PE=mgh$; Conservation of Energy
- Properties of Matter: Elasticity (Hooke's law, Young's modulus, bulk/shear modulus); Surface tension ($T=F/L$); Viscosity (η, Stokes' law)
- Waves & SHM: $v=f\lambda$; SHM: $a=-\omega^2x$, $T=2\pi\sqrt{(l/g)}$ for pendulum
- Rotational Motion: $\tau=I\alpha$; $L=I\omega$; Conservation of angular momentum L=constant
- Heat & Temperature: Modes — Conduction, Convection, Radiation; Stefan's law; Newton's law of cooling; Thermometry
- Geometric Optics: Mirrors & lenses ($1/f=1/v-1/u$); Refraction; Snell's law; Optical instruments
- Electrostatics: Coulomb's law $F=kq1q2/r^2$; $E=F/q$; $V=kq/r$; Capacitors $C=Q/V$; dielectrics
- Modern Physics: Laser (stimulated emission, applications); Superconductivity (zero resistance below Tc); Conventional vs Non-conventional energy sources

a-3 CHEMISTRY (3 Questions | 3 Marks)

- Water Hardness: Temporary (Ca/Mg bicarbonates) vs Permanent (sulfates/chlorides); Disadvantages; Removal: Boiling, Clark's, Zeolite/Ion Exchange, RO
- Acidity & Basicity: Arrhenius, Brønsted–Lowry, Lewis theory
- pH Scale: pH = –log[H■]; pH<7 acidic, pH=7 neutral, pH>7 basic
- Equivalent Weight: Wt of substance per mole of H■/OH■ involved; Eq.wt = Mol.wt / n-factor
- Electronic Configuration: Rules — Aufbau, Pauli, Hund's; config up to at. no. 30 (Zn)
- Symbol, Formula, Valency, Chemical Equation: balancing, types of reactions

a-4 COMMUNICATION SKILLS (3 Questions | 3 Marks)

- Synonyms & Antonyms: common word pairs
- Prepositions: in/on/at/by/for/with/to/from — usage rules
- Tenses: 12 tenses — structure & usage (Present/Past/Future × Simple/Continuous/Perfect/Perfect-Continuous)
- Voice: Active ↔ Passive conversion rules per tense
- Narration: Direct ↔ Indirect speech — reporting verbs, backshift of tenses, pronoun changes
- Punctuation: full stop, comma, semi-colon, colon, apostrophe, question mark
- Sentence Correction: subject-verb agreement, dangling modifiers, redundancy

a-5 GENERAL AWARENESS (3 Questions | 3 Marks)

- Haryana: 22 Districts, 72 Tehsils, 6,848 villages; Area = 44,212 km²; Pop ≈ 2.54 crore (2011); Borders: Punjab, Himachal Pradesh, UP, Rajasthan, Delhi (UT)
- LEET Eligibility: 3-yr Diploma in Engg./Tech. from recognized board with ≥45% (40% SC/ST) → direct 2nd year B.Tech.
- Technical Education Haryana: HSTES conducts LEET; Institutions offering B.Tech across Haryana (Govt. + Private SFI); B.Pharmacy institutions
- HSBTE Polytechnics: Haryana State Board of Technical Education; Govt. & aided polytechnics spread across state

- Online Counselling Steps: Registration → Document upload → Choice filling → Seat allotment → Document verification → Fee payment → Admission
- History of Haryana: Formed 1 Nov 1966 (carved from Punjab); Chandigarh is joint capital; Battle of Panipat fought on Haryana soil

★ No Negative Marking | 90 Minutes Total | 90 Questions | 90 Marks